ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	The energy left is too little to support further levels.
2	**(d)** 5000 J	5 x 10 x 10 x 10.
3	**(b)** UV radiation on oxygen	O2 splits and recombines.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Decomposition recycles nutrients.
5	**(a)** 40 kJ	Two transfers.
6	**(b)** lower levels have more energy	Less loss in the chain.
7	**(d)** A is false but R is true.	A is false: energy flows in one direction; R is true.
8	**(b)** insects and eagles	Insects (T2) and eagle (T5).
9	**(b)** Biodegradable waste	It is easily decomposed.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Biomagnification.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Grass = 50 x 100 = 5000 J (1); Snake = 5 J (1).	1+1
12	Energy decreases to 10^-9 of the original at the 10th level (1); Too little energy to sustain organisms (1).	1+1
13	Crop field has few species and needs inputs (1); Forest has many species and is self-sustaining (1).	1+1
14	Fox (T3) 80 J (1); Wolf (T4) = 8 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Small fish: 0.2 ppm (1); Big fish: 2 ppm (1); Kingfisher: 20 ppm (1).	1+1+1
16	Use both sides of paper (reduce) (1); Reuse bottles and bags (1); Recycle paper and glass; helps to conserve resources (1).	1+1+1
17	Toxic chemicals in effluents (1); Reduce oxygen levels (1); Poisoned food chain (1).	1+1+1
	OR: Food web has several connections (1); More predators or competitors for food (1); Chain reaction changes populations (1).	1+1+1
18	Releases toxic gases (1); Pollutes the air and harms organisms (1); Disturbs the food chains (1).	1+1+1
	OR: Vegetable waste decays (1); Plastic remains almost unchanged (1); Microbes decompose biodegradable matter only (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Zooplankton: 200,000 J (1); Small fish: 20,000 J (1); Large fish: 2000 J; birds: 200 J (1); Energy lost at the first transfer = 1,800,000 J (1); The first transfer has the largest loss (1).	1+1+1+1+1
	OR: Pesticide kills insects and remains in their bodies (1); Birds that eat insects accumulate it (1); Biomagnification poisons the birds (1); Alternative: biological control with natural predators (1); Alternative: crop rotation or organic farming (1).	1+1+1+1+1
20	Energy available decreases by 90% at each step (1); Less energy can support less biomass (1); Top carnivores are few in number (1); They need a large area to find food (1); They are the most affected by loss of energy at lower levels (1).	1+1+1+1+1
	OR: Cereals: humans at the second level (1); Cattle: humans at the third level (1); If plants hold 10,000 J, humans eating cereals get 1000 J (1); Via cattle: cattle 1000 J, humans 100 J (1); So cereals feed about 10 times as many (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Grass, goat, human	Fewer levels lose less energy.
2	**(b)** protect the ozone layer	Chlorine is avoided.
3	**(a)** Both A and R are true and R is the correct explanation of A.	UV absorption is the reason.
4	**(b)** grass increases and carnivores decrease	Fewer plant-eaters and less prey.
5	**(d)** Composting	Gives manure.
6	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R is not an explanation of A.
7	**(b)** 1000 J	Three transfers.
8	**(a)** using paper bags and reusing bottles	Protects environment.
9	**(d)** A is false but R is true.	A is false; R is true.
10	**(c)** Dumping of pesticide-laced waste	Pollution.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They convert dead organisms into simple inorganic substances (1); These are used again by plants (1).	1+1
12	Composting gives manure (1); Burning pollutes the air (1).	1+1
13	Pesticides get biomagnified (1); Top consumers hold the greatest concentration (1).	1+1
14	Producers: 10,000 J (1); Herbivores: 1000 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Forest: many species, complex food web (1); Crop field: few species, simple chain (1); Crop field is less stable and needs human inputs (1).	1+1+1
16	It is non-biodegradable (1); It persists in soil and water (1); It remains in food chains (1).	1+1+1
17	Herbivores would increase (1); Grass would be overgrazed (1); Eventually herbivores would starve (1).	1+1+1
	OR: They act on dead remains of all levels (1); They return nutrients to the soil (1); Producers need these nutrients (1).	1+1+1
18	It varies from organism to organism (1); Depends on the efficiency of use (1); But generally a small fraction is transferred (1).	1+1+1
	OR: Energy flows in one direction (1); It is lost as heat at each step (1); Materials, not energy, are recycled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Water to plankton: 10 times (1); Plankton to small fish: 10 times (1); Small fish to large fish: 10 times (1); Large fish to gulls: 10 times (1); Humans at high level accumulate harmful quantity (1).	1+1+1+1+1
	OR: Producers bring in energy and fix carbon (1); Decomposers recycle nutrients from dead matter (1); Energy flows one way and materials cycle (1); Without producers, no energy enters (1); Without decomposers, nutrients are locked in dead matter (1).	1+1+1+1+1
20	CFCs rise to the stratosphere (1); UV breaks CFCs and releases chlorine (1); Chlorine breaks down ozone to oxygen (1); Consequences: skin cancer, cataract, crop damage (1); Measures: UNEP agreement in 1987; individuals avoid CFC-based goods and aerosols (1).	1+1+1+1+1
	OR: (a) Toxic gases; remedy: recycle (1); (b) Ozone depletion; remedy: CFC-free products (1); (c) Water pollution, oxygen loss; remedy: sewage treatment (1); (d) Biomagnification; remedy: organic methods (1); Every action disturbs food chains (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** lose their food source	Dependent on the primary consumers.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Photosynthesis.
3	**(a)** biomagnification	Concentration increases up the chain.
4	**(b)** a fish-eating bird	Top.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Alternative paths.
6	**(b)** They can act at every trophic level on dead matter	They break down remains from every level.
7	**(b)** persist and accumulate	Remain in the environment.
8	**(a)** Both A and R are true and R is the correct explanation of A.	10 per cent law.
9	**(a)** Producers, herbivores, carnivores	Energy decreases at each higher trophic level.
10	**(d)** Crop field maintained by farmers	Man-made and managed.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They run off into water bodies (1); They enter food chains and biomagnify (1).	1+1
12	More UV reaches the surface (1); Harms phytoplankton, the producers, and the food chains (1).	1+1
13	Part is used in life processes like respiration (1); Part is lost as heat, so only 10% is passed on (1).	1+1
14	Many contain CFCs that harm ozone (1); They pollute the air (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(a) Many alternative pathways (1); (b) It biomagnifies and harms birds and humans (1); (c) Energy is not recycled and is lost as heat (1).	1+1+1
16	More demand for food (1); Plants and animals used excessively (1); Disturbs the balance and increases waste (1).	1+1+1
17	Producers are lost (1); The food chains that depend on them collapse (1); Soil is washed away and habitats are lost (1).	1+1+1
	OR: Very cold polar clouds help CFC reactions (1); Chlorine released destroys ozone (1); Sunlight in spring triggers the reaction (1).	1+1+1
18	Use cloth bags: no polythene waste (1); Use steel or glass bottles: reuse (1); Buy loose items: less packaging waste (1).	1+1+1
	OR: Organic waste is decomposed using oxygen (1); Oxygen level falls (1); Fish and other organisms die (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Segregation into dry and wet waste (1); Compost pits for wet waste (1); Recycling of paper, plastic, glass (1); Safe disposal of hazardous waste (1); Benefits: less pollution, manure, resource savings (1).	1+1+1+1+1
	OR: Forest: complex web with many species (1); Crop field: simple, few species (1); Forest recovers by alternate pathways (1); Crop field is disturbed more strongly (1); Crop field needs human care (1).	1+1+1+1+1
20	Initially herbivores increase (1); Plants are overgrazed (1); Food becomes scarce and herbivores starve (1); Disease and competition rise (1); So the claim is false; the balance is vital (1).	1+1+1+1+1
	OR: Producers: trees (1); Consumers: deer, rabbits, tigers, snakes; decomposers: fungi, bacteria (1); Chains: tree, deer, tiger and tree, rabbit, snake (1); Without decomposers dead matter and wastes would pile up (1); Nutrients would not be recycled and the producers would suffer (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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